

Quiz details

Quiz instructions:

This quiz will focus on the material from [Week 11](#) in the lab workbook. No data are needed to answer these questions.

☒ Show question details

Question 1 1 pts

Suppose that you wanted to use a randomisation approach to test whether or not fig wasp body volume different between 2 species. What would be one potential advantage of this approach?

- ☐ Randomisation would be faster than a t-test
- ☐ The assumptions of randomisation can be more easily generalised to the whole population
- ☐ Only the randomisation approach can be used to calculate confidence intervals
- ☐ The randomisation approach makes fewer assumptions about the data

Correct answer comments

[Section 34.4](#)  (https://bradduthie.github.io/SCIU4T4/Chapter_34.html#randomisation-assumptions) of the workbook noted that randomisation approaches make fewer assumptions about the underlying data than traditional parametric approaches.

Wrong answer comments

[Section 34.4](#)  (https://bradduthie.github.io/SCIU4T4/Chapter_34.html#randomisation-assumptions) of the workbook noted that randomisation approaches make fewer assumptions about the underlying data than traditional parametric approaches.

Question 2 1 pts

Assume that you wanted to use a randomisation approach to test the null hypothesis that a fig wasp's ovipositor length does not differ between 2 fig wasp species. You calculate the difference between mean species ovipositor length. You then randomise group identity (i.e., species 1 versus species 2) 999 times and calculate the difference between mean species ovipositor length with each randomisation. You find that the magnitude of the difference between randomised species ovipositor length exceeds that of the actual difference 57 times. What is the best unbiased estimate of your p-value? Report your answer to 3 or more decimal places.

0.058 (with margin: 0)

Correct answer comments

To answer this question, we need to take the number of mean difference from the randomisation that are more extreme than the actual data (57), plus 1 for the observation (1), then divide by the 999 randomisations plus the 1 observed,

$$P = \frac{57+1}{9999+1}.$$

The answer to the above is 0.058. For more explanation see the lab workbook [Chapter 34.3](https://bradduthie.github.io/SCIU4T4/Chapter_34.html#randomisation-for-hypothesis-testing) (https://bradduthie.github.io/SCIU4T4/Chapter_34.html#randomisation-for-hypothesis-testing).

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Question 3 1 pts

What kind of statistical test could you use to determine whether or not two groups have the same mean (check all that apply)?

- ☐ Linear regression
- ☐ Spearman rank correlation
- ☐ Independent samples t-test
- ☐ Randomisation test
- ☐ ANOVA
- ☐ Chi-square test

Correct answer comments

You could potentially use an Independent samples t-test (see [Chapter 21.2](https://bradduthie.github.io/SCIU4T4/Chapter_21.html#independent-samples-t-test) (https://bradduthie.github.io/SCIU4T4/Chapter_21.html#independent-samples-t-test)), a randomisation test (see [Chapter 34.2](https://bradduthie.github.io/SCIU4T4/Chapter_34.html#randomisation-for-hypothesis-testing) (https://bradduthie.github.io/SCIU4T4/Chapter_34.html#randomisation-for-hypothesis-testing)), or even a one-way ANOVA (see [Chapter 27.1](https://bradduthie.github.io/SCIU4T4/Chapter_27.html#one-way-anova-site) (https://bradduthie.github.io/SCIU4T4/Chapter_27.html#one-way-anova-site)) to test whether or not 2 groups have the same mean.

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Question 4 1 pts

Suppose that you bootstrapped means 999 times for the following values: 10.3, 11.4,

9.4, 13.5, 16.4, 11.0, 10.2, 12.3, 14.5, 13.4, 11.9, 12.6. What is the lowest *possible* mean value that you could get for any single bootstrap?

9.4 (with margin: 0)

Correct answer comments

In theory, it is possible to get a sample mean of 9.4, but only if this value was the only value sampled with replacement. This is highly unlikely, but possible. If we sample 12 new values with replacement, the probability that all 12 sampled would be the same value 9.4 would be $P = \left(\frac{1}{12}\right)^{12} \approx 0.0000000000001122$. That is, extremely unlikely! But this should make sense if the concept of what the bootstrap is doing is clearly understood.

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Question 5 1 pts

Which of the following randomisation approaches does not derive the null distribution from existing data?

- ☐ Resampling
- ☐ Bootstrapping
- ☐ Monte Carlo
- ☐ None of the above

Correct answer comments

See [Chapter 34.6](https://bradduthie.github.io/SCIU4T4/Chapter_34.html#monte-carlo)  (https://bradduthie.github.io/SCIU4T4/Chapter_34.html#monte-carlo) in the workbook, which explains the Monte Carlo procedure for generating a null distribution without resampling existing data.

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